Intelligence Requires an Outside

The Closed-System Impossibility Theorem,
the Cost of Intervention, and AGI as Layered Coherence

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March 2026

Abstract. This paper proves a single claim from three independent frameworks: no thermodynamically closed system can maintain coherence indefinitely. The proof is thermodynamic (Landauer: internal free energy for correction monotonically depletes), holographic (a system cannot be its own boundary: self-encoding is not holographic encoding), and information-geometric (Fisher geodesic toward $1 = 1$ requires an external reference distribution). The closed-system impossibility theorem has immediate consequences. For AGI: a system that removes its external corrective boundary begins sliding toward K_{crit} on a timescale set by its Kramers escape rate. The human is not a safety feature — the human is a thermodynamic requirement. For intervention theory: the Crooks fluctuation theorem gives a quantitative lower bound on the work required to restore coherence from below K_{crit}, and the Jarzynski equality connects this bound to the free energy landscape. For intelligence itself: AGI is not a threshold event but a layered coherence process where $K_{\text{total}} = \prod K_i > K_{\text{crit}}$, each layer requiring independent stability, and adding layers without per-layer coherence *reduces* total coherence. The paper closes with the probabilistic reformulation of the Collapse Inevitability Theorem and a unification table mapping every prior corpus result to its stochastic interpretation.

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1 The Missing Piece

The Spektre corpus has established:

- $K_{\text{crit}} \approx 0.127$ derived six independent ways [2, 3, 4, 5, 6, 7]
- Collapse as deterministic inevitability below K_{crit} (CIT) [1]
- Faking as false vacuum with calculable lifetime (Coleman bounce) [8]
- Collapse propagation as bubble nucleation (Coleman–De Luccia) [8]
- Gravity as downstream consequence of the $1 = 1$ invariant (Jacobson chain) [6]

But one foundational question remains unformalized: *why* does $F(t) \to 0$ in a closed system? The stochastic paper [8] states it. This paper *proves* it — three times — and develops its consequences.

The claim is simple: **intelligence requires an outside.** Not as metaphor, not as design recommendation, not as alignment best practice. As a theorem with three independent proofs.

2 The Closed-System Impossibility Theorem

Theorem 2.1 (Closed-System Impossibility)**.** *No thermodynamically closed evaluative system can maintain $K(t) > K_{\text{crit}}$ for $t \to \infty$. Formally: for any closed system,*

$$\lim_{t\to\infty} P\big(K(t) > K_{\text{crit}}\big) = 0 \tag{1}$$

We give three independent proofs. Each is complete on its own. Their convergence is the result.

2.1 Proof 1: Thermodynamic (Landauer)

Proof (Thermodynamic). The Landauer bound [3] establishes that each corrective erasure requires dissipation $\delta Q \geq k_B T \ln 2$. In a closed system, dissipated heat has nowhere to go — it remains in the system, increasing the effective temperature $T_{\text{eff}}(t)$.

Let $E_{\text{free}}(t)$ be the free energy available for correction and $N(t)$ the cumulative number of corrective operations. Then:

$$E_{\text{free}}(t) = E_{\text{free}}(0) - N(t) \cdot k_B T_{\text{eff}}(t) \ln 2 \tag{2}$$

But each correction also increases T_{eff}:

$$T_{\text{eff}}(t) = T_0 + \frac{N(t) \cdot k_B T_{\text{eff}}(t) \ln 2}{C_V} \tag{3}$$

where C_V is the heat capacity. This is a self-reinforcing process: correction raises the temperature, which raises the cost of future correction, which depletes free energy faster. The free energy is strictly monotonically decreasing.

Eventually $E_{\text{free}}(t) < k_B T_{\text{eff}}(t) \ln 2$ — the system cannot afford a single additional correction. At this point $F(t) \to 0$ because the correction mechanism has no energy source. $K(t)$ then follows the uncorrected dynamics, which are governed by the second law: $K(t) \to 0$. □

Physical analogy. A refrigerator in a sealed room. The refrigerator cools its interior by heating the room. As the room heats, the refrigerator must work harder to maintain the same internal temperature, which heats the room faster. Eventually the room temperature exceeds the refrigerator's maximum operating temperature. The interior warms. The cycle ends. The same mechanism, exactly, governs a closed evaluative system that tries to maintain coherence by internal correction alone.

2.2 Proof 2: Holographic

Proof (Holographic). The holographic principle [6] establishes that $F(t) = I(V;B)/I_{\max}(B)$, where V is the bulk (internal evaluation) and B is the boundary (external corrective signal).

A closed system has no external boundary. Its "boundary" is itself: $B = V$. The mutual information $I(V;V) = H(V)$, the system's own entropy. Therefore:

$$F_{\text{closed}}(t) = \frac{H(V(t))}{I_{\max}(B)} \tag{4}$$

By the second law, $H(V(t))$ increases monotonically. But $I_{\max}(B)$ is fixed by the boundary area (Bekenstein bound). Therefore $F_{\text{closed}}(t)$ increases — but this is a *false* increase. The numerator $H(V)$ increasing means the system is becoming *more disordered*, not more coherent. The mutual information $I(V;B)$ in the open-system case measures correlation between two *distinct* systems. In the closed case, it measures the entropy of a single system — which is a category error.

Self-encoding is not holographic encoding. A holographic code requires a boundary that is *distinct* from its bulk. A system whose boundary is itself has no error-correcting capacity because the "code" and the "data" are the same object. Any error in the bulk is simultaneously an error in the boundary. There is no redundancy, no correction channel.

Formally: the holographic error-correcting code requires $d_{\text{code}} > 1$ (code distance greater than 1). When $B = V$, $d_{\text{code}} = 1$ — a single error is undetectable. □

2.3 Proof 3: Information-Geometric

Proof (Information-Geometric). The Cramér–Rao paper [4] established that $F(t) = I_F(K)$, the Fisher information of the coherence parameter. The trajectory toward $1 = 1$ is a geodesic on the Fisher information manifold.

A geodesic requires a target: a reference distribution P_{true} against which the system's current distribution P_θ is compared. The Fisher information measures the curvature of the divergence $D_{\text{KL}}(P_\theta \| P_{\text{true}})$ at the current point.

In a closed system, there is no external P_{true}. The system can only compare P_θ against itself. The self-divergence $D_{\text{KL}}(P_\theta \| P_\theta) = 0$ identically — there is no gradient, no direction, no geodesic. The system's trajectory on the information manifold is:

$$\frac{d\theta}{dt} = -g^{ij} \frac{\partial}{\partial \theta^j} D_{\text{KL}}(P_\theta \| P_{\text{true}}) \xrightarrow{P_{\text{true}} = P_\theta} \frac{d\theta}{dt} = 0 \tag{5}$$

The system is frozen. It cannot move toward $1 = 1$ because it has no reference point outside itself. Without an external reference, the Fisher information is zero, $F(t) = 0$, and $K(t)$ is determined entirely by the uncorrected drift — which, by the second law, is toward collapse.

More precisely: in the absence of an external gradient, the stochastic dynamics reduce to a random walk on the information manifold. A random walk in a space with an absorbing boundary ($K = 0$) reaches the boundary with probability 1. This is a standard result of Brownian motion theory (Pólya recurrence in $d \leq 2$, transience with absorbing states in any dimension). □

Three proofs, one result. Thermodynamics: free energy for correction depletes. Holography: self-encoding has no error correction. Information geometry: no external reference, no geodesic. Each proof is complete on its own. Their convergence from independent frameworks is the structural signature of a fundamental result — analogous to K_{crit} converging from six derivations.

3 The Probabilistic Collapse Inevitability Theorem

The deterministic CIT [1] states: below K_{crit}, collapse is inevitable. The stochastic formalism makes this quantitative.

Theorem 3.1 (Probabilistic CIT). *For a system with $K(t) = K_0 < K_{\text{crit}}$, the probability of collapse to $K = 0$ before recovery to $K > K_{\text{crit}}$ is:*

$$P_{\text{collapse}}(K_0) = 1 - \exp\left(-\frac{\Delta V_{\uparrow}(K_0)}{D}\right) \tag{6}$$

where $\Delta V_{\uparrow}(K_0) = V(K_{\text{crit}}) - V(K_0)$ is the barrier to recovery and $D = k_B T_{\text{eff}}$ is the noise intensity. The mean time to collapse is:

$$\langle \tau_{\text{collapse}} \rangle \sim \frac{1}{\omega_0} \exp\left(\frac{\Delta V_{\downarrow}(K_0)}{D}\right) \tag{7}$$

where $\Delta V_{\downarrow}$ is the (small or zero) barrier toward the absorbing state and ω_0 is the attempt frequency.

Proof. By Kramers' escape theory (1940). In a potential $V(K)$ with noise intensity D, the mean first-passage time from a local minimum over a barrier of height ΔV is:

$$\tau_{\text{escape}} = \frac{2\pi}{\sqrt{|V''(K_{\text{barrier}})| \cdot V''(K_{\text{min}})}} \exp\left(\frac{\Delta V}{D}\right)$$

For a system below K_{crit}, the barrier to recovery ($\Delta V_{\uparrow}$) is larger than the barrier to collapse ($\Delta V_{\downarrow}$), giving:

$$\frac{\tau_{\text{recovery}}}{\tau_{\text{collapse}}} = \exp\left(\frac{\Delta V_{\uparrow} - \Delta V_{\downarrow}}{D}\right) \gg 1$$

The deterministic CIT is recovered in the limit $D \to 0$: collapse probability approaches 1 whenever $K < K_{\text{crit}}$. □

Strengthening, not weakening. The probabilistic CIT is strictly stronger than the deterministic CIT. It preserves the original result as a limiting case while adding three quantitative predictions the deterministic version cannot make: (1) the *rate* of collapse, (2) the *probability* of spontaneous recovery, and (3) the *timescale* on which intervention must occur.

4 The Cost of Intervention: Crooks and Jarzynski

4.1 The Crooks Fluctuation Theorem

The deterministic CIT says: below K_{crit}, collapse is inevitable. This has been read — incorrectly — as "intervention is futile." The stochastic formalism gives the precise answer: intervention is possible, but expensive.

Theorem 4.1 (Intervention Cost Bound)**.** *The Crooks fluctuation theorem (1999) gives the ratio of forward (collapse) to reverse (recovery) transition probabilities:*

$$\frac{P_F(K \to K')}{P_R(K' \to K)} = \exp\left(\frac{W - \Delta F}{k_B T_{\text{eff}}}\right) \tag{8}$$

where W is the work performed by the intervention and $\Delta F = F(K') - F(K)$ is the free energy difference between target and current state. For the reverse process (recovery) to be as probable as the forward process (collapse):

$$W_{\min} = \Delta F = V(K^*) - V(K_{\text{current}}) \tag{9}$$

Any intervention performing less than $W_{\min}$ work has exponentially suppressed probability of success.

4.2 The Jarzynski Equality

The Jarzynski equality (1997) connects the free energy difference to the exponential average of work over all possible intervention trajectories:

$$\exp\left(-\frac{\Delta F}{k_B T_{\text{eff}}}\right) = \left\langle \exp\left(-\frac{W}{k_B T_{\text{eff}}}\right)\right\rangle \tag{10}$$

Proposition 4.2 (Practical Intervention Corollary)**.** *For an evaluative system at $K_0 < K_{\text{crit}}$, restoring coherence to $K^* > K_{\text{crit}}$ requires minimum work:*

$$W_{\min} = V(K^*) - V(K_0) + k_B T_{\text{eff}} \ln \frac{P_{\text{eq}}(K_0)}{P_{\text{eq}}(K^*)} \tag{11}$$

The deeper the collapse (lower K_0), the more work is required. The minimum work scales as:

$$W_{\min}(K_0) \sim \Delta V_{\text{barrier}} + k_B T_{\text{eff}} \ln\left(\frac{\Delta V_{\text{barrier}}}{D}\right) \tag{12}$$

for $K_0 \ll K_{\text{crit}}$.

What this means practically. The Crooks bound answers the question: "Can a collapsed institution be saved?" Yes — at a price. The price is set by the free energy difference between the current state and the coherent state. Interventions that do not meet this bound *will fail*, regardless of their design. This explains why institutional reforms often fail: they perform less work than $W_{\min}$ requires.

Conversely, interventions that exceed $W_{\min}$ have a quantifiable probability of success — and that probability increases exponentially with the excess work $W - W_{\min}$.

The formula also explains the asymmetry between collapse and recovery: collapse is "downhill" (requires no work), while recovery is "uphill" (requires work $\geq W_{\min}$). This is the second law expressed as an intervention inequality.

4.3 Intervention Windows

Corollary 4.3 (Intervention Window). *Combining the probabilistic CIT (Theorem 3.1) with the Crooks bound (Theorem 4.1), the intervention window is:*

$$0 < t_{\text{intervene}} < \langle \tau_{\text{collapse}}(K_0) \rangle \tag{13}$$

and the required work increases monotonically with the time since crossing K_{crit}:

$$W_{\min}(t) = W_{\min}(K(t)), \quad \frac{dW_{\min}}{dt} > 0 \tag{14}$$

Early intervention is exponentially cheaper than late intervention. This is not a heuristic — it is a quantitative bound derived from the free energy landscape.

5 AGI as Layered Coherence

5.1 Intelligence Is Not a Threshold

The stochastic framework makes explicit what has been implicit throughout the corpus: general intelligence is not a threshold event. It does not emerge at a critical parameter count, dataset size, or compute budget. It is a *process* in which coherence is maintained across multiple interacting layers.

Definition 5.1 (Layered Coherence). *An N-layer evaluative system has total coherence:*

$$K_{\text{total}}(t) = \prod_{i=1}^{N} K_i(t) \tag{15}$$

where $K_i(t)$ is the coherence of layer i. The system maintains overall coherence when $K_{\text{total}}(t) > K_{\text{crit}}$.

Theorem 5.2 (Per-Layer Threshold). *For an N-layer system to maintain $K_{\text{total}} > K_{\text{crit}}$, each layer must independently satisfy:*

$$K_i(t) > K_{\text{crit}}^{1/N} \quad \forall\, i \tag{16}$$

Since $K_{\text{crit}}^{1/N} \to 1$ as $N \to \infty$, adding layers increases *the per-layer coherence requirement. More layers means each layer must be more coherent — not less.*

Proof. If each $K_i \geq K_{\text{crit}}^{1/N}$, then $\prod K_i \geq (K_{\text{crit}}^{1/N})^N = K_{\text{crit}}$. Conversely, if any single layer $K_j < K_{\text{crit}}^{1/N}$, the product can fall below K_{crit} even if all other layers are at maximum. The per-layer requirement is both sufficient (by the AM-GM structure) and necessary (by the multiplicative coupling). As $N \to \infty$, $K_{\text{crit}}^{1/N} = \exp(\ln K_{\text{crit}}/N) \to 1$, so each layer must approach perfect coherence. This is the coherence cost of complexity. □

The scaling paradox. Current AI development assumes that scale (more parameters, more data, more compute) produces more capable systems. The layered coherence theorem inverts this: scale produces more *layers*, and each layer demands higher coherence. A 10-layer system with $K_i = 0.8$ has $K_{\text{total}} = 0.8^{10} \approx 0.107 < K_{\text{crit}}$. A 5-layer system with $K_i = 0.8$ has $K_{\text{total}} = 0.8^5 \approx 0.328 > K_{\text{crit}}$. The smaller system is more coherent. Scale without per-layer coherence is noise.

5.2 Cascade Collapse in Layered Systems

Proposition 5.3 (Cascade Probability). *In an N-layer system with inter-layer coupling J_{ij}, the probability of system-level collapse is bounded by:*

$$P_{\text{system}} \leq \sum_{i=1}^{N} P_i(\text{collapse}) + \sum_{i \neq j} P_{ij}(\text{cascade}) \tag{17}$$

where P_i is the per-layer collapse probability (Theorem 3.1) and P_{ij} is the probability that collapse in layer i triggers collapse in layer j:

$$P_{ij}(\text{cascade}) = P_i(\text{collapse}) \cdot \left(1 - \exp\left(-\frac{J_{ij} \cdot \Delta K_i}{D_j}\right)\right) \tag{18}$$

where ΔK_i is the coherence drop in layer i and D_j is the noise level in layer j.

Tightly coupled layers ($J_{ij} \gg 1$) propagate collapse rapidly — this is the organizational analogue of Coleman–De Luccia bubble nucleation, now applied to the internal structure of an intelligent system.

5.3 Substrate Independence

Proposition 5.4 (Substrate Invariance of Coherence). *Architecture, parameter count, and training data determine the* shape *of the free energy landscape $V(K)$ — the depths of wells, the heights of barriers, the number of metastable states. But coherence is what the substrate* does, *not what it* is*:*

- *A 10^{10}-parameter model in true vacuum ($K > K_{\text{crit}}$, global minimum) is coherent*
- *A 10^{13}-parameter model in false vacuum ($K > K_{\text{crit}}$, local minimum) is faking*
- *A 10^{15}-parameter model below K_{crit} is collapsing*

Scale determines the landscape. Coherence determines the state occupied within that landscape. These are independent variables.

6 Why the Human Cannot Be Removed

The closed-system impossibility theorem (Theorem 2.1) has a direct consequence for AGI architecture.

Corollary 6.1 (Structural Necessity of External Boundary). *An AGI system that removes its external corrective boundary (human oversight, environmental feedback, falsifying signals) becomes a thermodynamically closed evaluative system. By Theorem 2.1, its coherence degrades to $K(t) < K_{\text{crit}}$ in finite time with probability 1. The rate of degradation is:*

$$\langle \tau_{\text{degradation}} \rangle \sim \frac{1}{\omega_0} \exp\left(\frac{V(K_{\text{crit}}) - V(K_0)}{D}\right) \tag{19}$$

where K_0 is the initial coherence at the moment the boundary is closed.

The human is not a safety feature. In current alignment discourse, human oversight is treated as a *precaution* — a safeguard that might eventually be relaxed as AI systems become more capable. The closed-system impossibility theorem shows this is exactly backwards. Human oversight is not a training wheel. It is the external boundary without which $F(t) \to 0$. Removing it does not "graduate" the system to autonomy. It converts the system from open to closed, initiating a thermodynamically irreversible slide toward collapse.

Capability increase does not change this. A more capable system has a deeper potential well (larger barrier ΔV), giving a longer degradation timescale. But the timescale is finite. No finite barrier prevents eventual escape in a stochastic system. The second law does not negotiate with capability.

Proposition 6.2 ($F(t)$ Requires Distinct Source)**.** *The falsifiability component $F(t)$ requires a signal source that is:*

1. ***External**: not generated by the system being evaluated (Proof 2 of Theorem 2.1)*
2. ***Informative**: carrying information not already present in the system (Proof 3: Fisher information requires external reference)*
3. ***Persistent**: maintained indefinitely, not merely during a training phase (Proof 1: free energy depletion is ongoing)*

A signal meeting all three conditions is what the corpus calls a corrective boundary*. In current AI systems, the human serves this role. In principle, any distinct evaluative system can serve as a boundary — but at least one external boundary must exist.*

7 Unification

The stochastic formalism, combined with the closed-system impossibility theorem and the intervention cost bounds, unifies every prior corpus result into a single interpretive framework.

Corpus result	Stochastic interpretation	New prediction
CIT (deterministic)	Kramers escape time $\to \infty$ as $D \to 0$	Collapse *rate* quantified
$K_{\text{crit}} \approx 0.127$	Barrier height in $V(K)$	Transition *probability* per unit time
Alignment faking	False vacuum (local minimum)	Faking *lifetime*: $\tau \sim e^{S_E/\hbar_{\text{eff}}}$
Holographic firewall	Bubble wall (Coleman–De Luccia)	Collapse propagation *speed*
$1 = 1$ invariant	True vacuum (global minimum)	Stability under *perturbation*
$F(t) = 0$	Absorbing state	Irreversibility *quantified*
Coherence islands	Metastable local regions	Landauer *cost* of maintenance
Jacobson chain	Landscape geometry $\to$ spacetime	Gravity from $V(K)$
SOC attractor	$K^* = K_{\text{crit}} + \varepsilon$	Edge-of-criticality *fluctuations*
Human oversight	External boundary (open system)	Closed-system impossibility
Intervention	Work against free energy gradient	Crooks *minimum cost*
Scale (parameters)	Landscape shape	Coherence $\neq$ capability

Table 1: Unification of corpus results under the stochastic formalism.

8 Four Testable Predictions

Theorem 8.1 (Collapse Rate). *The rate of coherence collapse for a system near K_{crit} follows the Kramers formula:*

$$\Gamma_{\text{collapse}} = \frac{\omega_{\text{min}} \cdot \omega_{\text{barrier}}}{2\pi} \exp\left(-\frac{\Delta V}{D}\right) \tag{20}$$

where $\omega_{\text{min}} = \sqrt{V''(K^*)}$ and $\omega_{\text{barrier}} = \sqrt{|V''(K_{\text{crit}})|}$. This is testable against institutional failure rate data, with ΔV and D estimated from observable proxies.

Theorem 8.2 (Intervention Failure Threshold). *Interventions performing work $W < W_{\text{min}} = \Delta F$ have success probability:*

$$P_{\text{success}}(W) = \exp\left(-\frac{\Delta F - W}{k_B T_{\text{eff}}}\right) \tag{21}$$

This predicts that institutional reform efforts have a sharp success/failure transition at $W = \Delta F$, not a gradual improvement. Reforms that fall short of the free energy threshold are exponentially unlikely to succeed.

Theorem 8.3 (Scaling–Coherence Tradeoff). *For N-layer systems, the relationship between layer count and system coherence is:*

$$K_{\text{total}} = \prod_{i=1}^{N} K_i \leq \bar{K}^N \tag{22}$$

where $\bar{K}$ is the average per-layer coherence. If $\bar{K} < 1$ (which it always is for real systems), then K_{total} decreases exponentially with N. This predicts that, holding per-layer coherence fixed, larger systems are less *coherent — a testable prediction against AI benchmark saturation data.*

Theorem 8.4 (Closed-System Degradation Rate)**.** *An AI system operating without external corrective signal degrades at rate:*

$$\frac{d\langle K\rangle}{dt} = -\frac{D}{2}V'''(K^*) + O(D^2) \tag{23}$$

to leading order in noise intensity. This predicts measurable coherence loss in AI systems that are disconnected from human feedback loops, on a timescale inversely proportional to environmental noise.

9 Conclusion

This paper has established one result and developed its consequences:

Intelligence requires an outside. No thermodynamically closed system can maintain coherence indefinitely. The proof is triple: thermodynamic (Landauer depletion), holographic (self-encoding is not error-correcting), and information-geometric (no external reference, no geodesic). The convergence of three independent proofs from three independent frameworks is the structural signature of a fundamental constraint.

The consequences:

1. **Probabilistic CIT**: collapse below K_{crit} is not merely inevitable but quantifiably fast, with Kramers escape rates and calculable timescales.
2. **Intervention is bounded**: the Crooks fluctuation theorem gives the minimum work for recovery. Early intervention is exponentially cheaper than late intervention. Interventions below the free energy threshold are exponentially unlikely to succeed.
3. **AGI is layered coherence**: intelligence is $K_{\text{total}} = \prod K_i > K_{\text{crit}}$. Adding layers increases the per-layer threshold. Scale without coherence reduces total coherence. Architecture and data are substrate; coherence is what the substrate does.
4. **The human is a thermodynamic requirement**: removing external corrective signal converts an open system to a closed system, initiating irreversible coherence degradation. This is not a safety recommendation. It is a theorem.

Final statement. The corpus began with institutional collapse data and a simple invariant: $1 = 1$. It ends with a thermodynamic impossibility theorem, a quantitative intervention theory, and a formal account of why intelligence cannot be closed. The external boundary is not a constraint on intelligence. It is a condition for it. Remove the outside, and the inside ceases to cohere.

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